Benchmarking LLM Evaluative Performance: Development of a Pipeline for Automatic Validation

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Abstract

The rapid advancement of large language models (LLMs) has positioned them as promising tools for automating evaluative tasks traditionally performed by humans, such as assessing text quality and benchmarking abstract constructs. Despite their potential, questions persist regarding the reliability, and validity of LLM-based evaluation, particularly when compared to human expert standards. This thesis addresses these concerns through the development and empirical validation of a modular, automatic evaluation pipeline focused on external validation of LLM judgments. The primary objective was to create a transparent, reproducible workflow—operationalized as the Copernicus system—that enables systematic, rubric-driven benchmarking of LLM evaluative performance. The pipeline integrates rubrics, ensemble LLM evaluation, and survey automation via the Qualtrics API, with participant recruitment and demographic balancing facilitated through Prolific. Empirical studies conducted using this pipeline demonstrate improved alignment of LLM-generated scores with human expert ratings, greater inter-rater reliability, and enhanced granularity compared to standard, one-shot LLM evaluations. The original contribution of this work is the development and implementation of a robust, extensible validation pipeline that supports high-throughput, externally benchmarked assessment of LLM evaluative outputs. This infrastructure advances best practices in quality control, and methodological transparency for automated AI assessment. Implications include the potential for more scalable and reliable evaluation of AI systems, with recommendations for continued integration of human benchmarks to address evolving reliability and bias concerns.

Keywords: large language models; evaluation pipeline; rubric-based assessment; external validation; automated benchmarking

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Chapter 1

Introduction

1.1 Background and Motivation

The rapid emergence of large language models (LLMs) has fundamentally transformed the landscape of natural language processing and artificial intelligence. LLMs, such as ChatGPT, Gemini, and Claude, have demonstrated unprecedented capabilities in generating coherent text, solving complex reasoning tasks, and simulating human like behaviors across a wide range of domains. With these advances, there is increasing interest in leveraging LLMs not only as tools for text generation but also as evaluators or "judges" of complex outputs a trend referred to in the literature as "LLM as a judge" or "LLM as an examiner" (Gu et al. 2024; Schroeder et al. 2024; Zheng et al. 2023).

Evaluative performance is central in many scientific, educational, and organizational workflows, where the ability to score, assess, or benchmark artifacts is critical for quality assurance, research, and innovation. However, traditional human evaluation is expensive, time consuming, and sometimes inconsistent, leading to the pursuit of scalable, reliable, and cost effective alternatives. LLMs offer the promise of automating this process, yet their use as evaluators brings a host of methodological, practical, and theoretical challenges (Boussioux et al. 2024; Doshi et al. 2024; Park et al. 2024).

One key motivation for this thesis is the growing need for external validation of LLM based evaluation workflows. While many benchmarking efforts focus on internal model

performance, the external reliability, validity, and robustness of LLM judgments remain open questions (Zheng et al. 2023). Empirical findings suggest that while LLMs can approximate human evaluation in aggregate sometimes outperforming non experts their outputs can exhibit inconsistency, bias, and susceptibility to prompt design (Doshi et al. 2024; Schroeder et al. 2024; Tian et al. 2024). Furthermore, recent studies have demonstrated that aggregation and diversity among evaluators (both human and artificial) can improve overall evaluation quality, echoing the "wisdom of crowds" effect in decision science (Doshi et al. 2024; Park et al. 2024).

Another motivation is the shift toward agentic AI workflows, which orchestrate multiple agents (human or AI) through structured pipelines for evaluation and validation. This paradigm supports external, modular validation of LLM outputs using multi stage, rubric based, and sometimes multi agent processes (Boussioux et al. 2024). In this context, the need for systematic, automatic validation pipelines capable of integrating rubrics, extracting and scoring textual evidence, and comparing LLM and human outputs has become increasingly acute for both researchers and practitioners.

1.2 Scope of the Thesis

This thesis addresses the challenge of benchmarking the evaluative performance of LLMs through the development and validation of an automatic evaluation pipeline. The scope of the work is defined along several key dimensions:

(i) **Focus on Evaluative Tasks:** The core focus is on tasks where LLMs are used to score, judge, or benchmark text based artifacts according to defined criteria or rubrics. This includes, but is not limited to, assessments in scientific reasoning, business model evaluation, creative problem solving, and social science benchmarking.

(ii) **Pipeline Development:** The thesis details the design, implementation, and empirical assessment of a modular pipeline for automatic validation, incorporating

state of the art practices in rubric construction, evidence extraction, and multi stage LLM workflows (e.g., as embodied in the Copernicus system) .

(iii) **External Validation:** Emphasis is placed on "external" validation that is, validation which goes beyond internal model agreement, benchmarking LLM outputs against human expert ratings and/or robust, multi agent reference standards (Doshi et al. 2024; Zheng et al. 2023).

(iv) **Rubric based Evaluation:** The pipeline leverages granular, literature informed rubrics and attribute/sub attribute structures, following best practices in rubric design for both human and AI assessment .

(v) **Exclusions:** The thesis does not focus on LLM training or fine tuning, low level model interpretability, or non textual evaluative domains. While it draws from broader LLM and evaluation literature, the primary empirical and methodological contributions are centered on the evaluation pipeline and benchmarking studies.

1.3 Structure of the Thesis

This thesis is organized as follows:

(i) **Chapter 2: Literature Review** surveys the key literature on agentic AI workflows, LLMs as evaluators and judges, synthetic data generation, and rubric based evaluation, highlighting gaps and open questions relevant to this research.

(ii) **Chapter 3: The Copernicus System** introduces the Copernicus measurement system, its rationale, and its role as an external validation platform, providing a summary of its architecture and primary functions.

(iii) **Chapter 4: The Need for External Validation** explores why external validation is essential for LLM evaluation, contrasting it with internal validation, and discusses key challenges and opportunities.

(iv) **Chapter 5: Pipeline Design** presents the design and rationale of the proposed automatic evaluation pipeline, including stepwise processes, trade offs in rubric and sampling strategies, and considerations of quality and reliability.

(v) **Chapter 6: Qualtrics Implementation** details the integration of the pipeline with Qualtrics for survey based evaluation, including automation via the Qualtrics API and key implementation steps.

(vi) **Chapter 7: Prolific Study Launch** describes the process of deploying the evaluation workflow in human studies, outlining participant criteria, remuneration, demographic considerations, and data monitoring.

(vii) **Chapter 8: Evaluation Methods and (AI) Agents** reviews the methodology for evaluating both LLM and human agent performance, and discusses reliability, validity, and statistical considerations.

(viii) **Chapter 9: Discussion** synthesizes the implications of the workflow, identifies limitations, and offers recommendations for future work.

(ix) **Chapter 10: Conclusion** summarizes the thesis' contributions and final remarks.

Chapter 2

Literature Review

2.1 Agentic AI Workflows

Agentic AI workflows represent a paradigm shift from traditional, tool-like applications of artificial intelligence toward integrated, autonomous, and interactive systems capable of multi-step reasoning, decision-making, and adaptation. These workflows are characterized by their capacity to orchestrate complex tasks, maintain conversational or task state, invoke external tools or APIs, and generate or validate content in an iterative, feedback-driven manner.

Recent advances in large language models (LLMs) and API ecosystems, such as OpenAI's Assistants API, exemplify this shift by enabling the creation of stateful, modular AI assistants that encapsulate not only a core language model, but also specialized instructions, tool access, retrieval abilities, and persistent memory of prior interactions. In this agentic model, the primitives shift from stateless completions to entities like *Assistants*, *Threads*, and *Runs*, reflecting a workflow-oriented design that mirrors human-like task management and learning over time.

In organizational and scientific contexts, agentic workflows have been leveraged to augment creative problem-solving, orchestrate synthetic data generation, and automate evaluation pipelines. For example, Boussioux et al. show that agentic human-AI collaborations, mediated by strategic prompt engineering and iterative refinement, can outper-

form purely crowd-based approaches in certain domains, generating solutions of higher strategic value and quality at lower cost and scale (Boussioux et al. 2024). This evidences the emerging power of "AI-in-the-loop" and workflow-centric design for advancing complex, open-ended tasks.

Moreover, agentic workflows facilitate the deployment of evaluative pipelines in research and industry, as seen in the Copernicus system, which enables custom measurement models for abstract constructs via agentic, multi-step evaluation processes that combine evidence extraction and rubric-based scoring.

2.2 LLMs as Evaluators and Judges

A growing body of literature investigates the use of large language models as evaluators, judges, or examiners, both in human-centered domains and for the assessment of AI-generated content. The "LLM-as-a-judge" paradigm, first articulated by Zheng et al., positions LLMs as scalable and explainable surrogates for human evaluators in a variety of tasks where human preference data is scarce or costly to obtain (Zheng et al. 2023). Subsequent work, such as Gu et al., expands this paradigm by introducing the concept of "LLM-as-examiner," where models apply Likert-style scales and granular rubrics to score artifacts, mimicking human grading practices.

Evaluation by LLMs hinges on three main strategies: (1) prompt design to elicit effective evaluation behavior, (2) improvement of LLM evaluation capabilities through fine-tuning or feedback, and (3) optimization of final results via aggregation, score smoothing, or transformation from pointwise to pairwise judgments. Prompt design has proven particularly critical, with few-shot prompting and the decomposition of evaluation steps and criteria (e.g., HD-eval framework) improving alignment and reliability of LLM-based judgments.

Despite progress, challenges remain. Schroeder and Wood-Doughty, for instance, highlight variability in LLM reliability depending on temperature and task, revealing trade-offs between reliability and performance. Tian et al. further show that LLM evaluation

is task-dependent, with rubric-based, example-rich prompts outperforming zero-shot approaches. Chain-of-thought (CoT) prompting and crowd-based comparative evaluation are also emerging as strategies to boost evaluative accuracy and interpretability (Zheng et al. 2023).

The use of LLMs as judges has also been explored in strategic decision-making contexts (Doshi et al. 2024), behavioral simulation (Park et al. 2024), and creative domains, demonstrating both the opportunities and the limitations of relying on artificial evaluators. Notably, aggregating predictions from diverse LLMs, roles, and prompts ("wisdom of the artificial crowd") has been shown to improve alignment with human expert rankings, underscoring the value of diversity and scale in LLM evaluation.

2.3 Synthetic Data Generation

Synthetic data generation using LLMs has become a powerful tool for augmenting datasets, simulating human responses. Recent studies have demonstrated that LLMs can replicate human survey responses and even predict election outcomes by conditioning on demographic and behavioral data (Jiang et al. 2025). For example, in social science, LLMs have been tasked with generating synthetic responses that match human participants' answers across multiple domains, with notable success in reproducing means and variances for a variety of items, although certain biases and reduced variability persist.

Further, Park et al. develop an agentic simulation framework in which LLM-based generative agents, grounded in qualitative interviews, replicate the attitudes and behaviors of over 1,000 real individuals. These agents achieve up to 85% of the self-consistency of human participants on standard social science instruments, providing a new foundation for simulating both individual and collective behaviors (Park et al. 2024).

In the context of benchmarking and evaluation, synthetic data enables scalable and controlled assessment of model capabilities. It is also increasingly used to supplement or replace human annotation in domains ranging from public opinion research to strategic decision analysis, although attention to prompt engineering, demographic conditioning,

and bias correction remains essential.

2.4 Rubric-Based Evaluations

Rubric-based evaluation schemes provide structure, transparency, and reproducibility to subjective assessment tasks. In the domain of LLM evaluation, rubric-based approaches operationalize multi-dimensional constructs—such as scientific reasoning, creativity, or argument quality—into concrete, graded criteria.

The Copernicus system exemplifies this approach, defining a set of core dimensions (e.g., Assumptions, Problem Formulation, Attribute Formulation, Logical Links, Data Orientation, Belief Update, Methodic Doubt) and further decomposing these into granular markers and subrubrics. Each marker is assessed via detailed Likert-type scales, with high inter-rater agreement achieved through precise definitions and structured evidence extraction.

The rubric development process follows best practices from both educational assessment and measurement science: (1) grounding in research definitions and practical measures, (2) defining independent, comprehensive attributes, (3) decomposing attributes into observable, text-based markers, (4) calibrating scale levels for each sub-concept, and (5) encoding rubrics for reproducible application. The marker and subrubric definitions are designed for both human and AI-based annotation, supporting hybrid and fully automated pipelines.

Empirical results indicate that rubric-based, multi-step evaluations can improve alignment with human expert judgment compared to standard "one-shot" LLM assessment approaches. Hierarchical decomposition and marker-based evidence extraction, as in the Copernicus and HD-eval frameworks, are particularly effective in enhancing explainability and reliability.

2.5 Relevant Papers and Resources

The literature on benchmarking LLM evaluative performance, agentic AI workflows, synthetic data, and rubric-based assessment is rapidly growing. Key contributions and resources relevant to this thesis include:

Park et al. (2024) introduce a novel architecture for generative agents, leveraging large language models (LLMs) to simulate the attitudes and behaviors of over 1,000 real individuals based on in-depth qualitative interviews. Their methodology centers on anchoring agent behavior in rich, individual-level data rather than demographic or summary descriptions, enabling a more nuanced and accurate replication of human responses across diverse domains, including canonical social science surveys (e.g., the General Social Survey), personality inventories, behavioral economic games, and experimental replications. The validation approach is rigorous: the simulated agents' responses are benchmarked against the actual responses of their human counterparts, with accuracy normalized by each participant's self-consistency over time. The results demonstrate that interview-informed agents can replicate participants' attitudes and behaviors nearly as well as the participants themselves can over a two-week interval, and significantly outperform agents constructed only from demographic or summary data. Notably, the study reports reduced accuracy biases across racial and ideological groups in interview-based agents, addressing concerns of demographic stereotyping.

Shao et al. (2024) present DeepSeekMath, a domain-specific large language model (LLM) designed for advanced mathematical reasoning, and their work is notable for its rigorous approach to LLM validation in highly structured, quantitative domains. The core validation methodology hinges on extensive benchmarking across both English and Chinese mathematical datasets, including GSM8K, MATH, MMLU-STEM, and others, utilizing both chain-of-thought and program-aided reasoning formats. Their evaluation employs standardized metrics such as top-1 accuracy and pass rates, directly comparing DeepSeekMath's performance to leading open- and closed-source models, notably outperforming other open models and approaching the performance of proprietary systems like GPT-4 and Gemini-Ultra. Crucially, the authors also introduce Group Rela-

tive Policy Optimization (GRPO), a memory-efficient reinforcement learning algorithm, and validate its effectiveness by showing significant performance gains when applied to instruction-tuned models, both in-domain and out-of-domain.

Jiang et al. (2025) provide a comprehensive examination of LLMs as synthetic evaluators of public opinion, focusing on survey and election prediction tasks. Their core methodology centers on the "Matching-LLM" approach, which integrates synthetic responses generated by ChatGPT-4o—conditioned on detailed demographic profiles—with historical human response data, using propensity score matching and optimized weighting to align LLM outputs more closely with real-world outcomes. The evaluation spans both in-sample (replicating known survey and election results) and out-of-sample predictions (notably, the 2024 U.S. presidential election), benchmarking LLM performance using metrics such as mean absolute deviation, response variance, and the preservation of pairwise correlations between survey items. Notably, the Matching-LLM method demonstrates superior alignment with human data compared to LLMs alone, effectively capturing cross-cultural differences and complex behavioral patterns.

Boussioux et al. (2024) rigorously investigate the evaluative performance of LLMs in creative problem-solving by benchmarking AI-generated and human-generated solutions within a controlled crowdsourcing experiment. Their methodology centers on a large-scale field experiment comparing solutions to circular economy challenges produced by (a) a human crowd, and (b) human-AI collaborations leveraging GPT-4, using strategically engineered prompts. Validation of LLM outputs is conducted through a blinded evaluation by 300 external human judges, employing rubric-based ratings along five axes: novelty, strategic viability, environmental value, financial value, and overall quality. Hierarchical linear models are used to analyze over 3,900 evaluator-solution pairs, quantifying both mean ratings and top-decile performance, thereby enabling robust external validation against human benchmarks.

Doshi et al. (2024) investigate the reliability of large language models (LLMs) as artificial evaluators in the domain of strategic decision making, focusing specifically on the evaluation of business models. Their validation methodology is structured around two

studies: one using AI-generated business models and another using real-world competition entries. In both, LLMs from multiple providers (OpenAI, Google, Meta, Anthropic) were prompted—under diverse roles and instructions—to rank business models via pairwise comparisons. The study introduces a systematic aggregation framework, measuring agreement between AI rankings and those of human experts using multiple metrics (Pearson and Spearman correlation, top and bottom choice agreement), and decomposing the value of aggregation into "diversity" (across models, roles, prompts) and "scaling" (number of predictions) effects.

Chapter 3

The Copernicus System

3.1 Overview and Rationale for Copernicus

The Copernicus system is a measurement tool specifically designed to quantify and analyze abstract and difficult-to-measure constructs, such as "scientificness" or "customer centricity," within any form of verbal communication, including written text and speech. Its development was motivated by the growing need among researchers and organizations to obtain scalable, reliable, and interpretable measures of qualitative phenomena that have traditionally eluded systematic quantification. Copernicus addresses this gap by enabling the agentic, AI-driven creation, customization, and deployment of measurement models, thereby transforming the way complex constructs are operationalized and evaluated in both research and practice.

A central rationale behind Copernicus is the demand for robust tools that can assess the adoption and impact of scientific reasoning and decision-making, particularly in managerial and organizational contexts. Existing methods for evaluating such reasoning are often labor-intensive, subjective, and inconsistent, hampering efforts to benchmark and improve decision quality at scale. By leveraging large language models (LLMs) and systematic rubric-based evaluations, Copernicus aims to provide transparent, reproducible, and high-throughput measurement of nuanced reasoning attributes.

3.2 Role of Copernicus in External Validation

Copernicus plays a pivotal role in the external validation pipeline for benchmarking LLM evaluative performance. In this context, external validation refers to the assessment of LLM or human-generated outputs using criteria or evaluators that are independent of the original system, thereby providing an objective standard against which internal performance metrics can be compared.

The system operationalizes external validation by:

(i) Translating high-level constructs (e.g., "scientific reasoning") into multi-dimensional, literature-based rubrics that are grounded in established research.

(ii) Enabling the consistent application of these rubrics across a wide array of textual data—ranging from scientific articles to business reports, using an ensemble of LLMs and models.

(iii) Integrating a unique two-step "markers" approach, wherein granular evidence for each rubric attribute is first extracted and then subjected to a secondary evaluative analysis. This decomposition increases the reliability and reduces the stochasticity associated with one-shot or prompt-based evaluations common in standard LLM workflows.

(iv) Supporting large-scale, comparative benchmarking studies that require transparent, repeatable, and explainable evaluation pipelines.

By establishing a formalized, rubric-driven framework for external evaluation, Copernicus not only facilitates the measurement of LLM alignment with human values or scientific standards but also provides a benchmark for ongoing improvements in agentic evaluation pipelines. The external validation results derived from Copernicus have been shown to align well with human expert ratings in studies across various domains, further supporting its value as a reference point for both research and applied settings.

3.3 Brief Summary of Architecture and Main Functions

Copernicus is architected as a modular, extensible pipeline that integrates several core components and functions:

1. Attribute and Rubric Definition

(i) Copernicus employs a structured process for defining evaluation rubrics, beginning with an extensive review of how key concepts are defined and measured in the literature.

(ii) Attributes (dimensions) are selected to be independent and broad, with detailed written descriptions and decomposition into sub-attributes or markers. These are made operational via granular, five-level Likert-style rubrics for each sub-concept, ensuring both breadth and specificity.

2. The "Markers" Two-Step Evaluation Process

1. **Evidence Extraction**: For each attribute (e.g., Assumptions, Problem Formulation), Copernicus identifies textual markers that signal the presence and quality of the attribute, using a detailed rubric. This step requires the extraction of explanations or evidence snippets corresponding to rubric levels.

2. **Evidence Evaluation**: The extracted evidence is reformatted as an analytical text and rescored using the master rubric for each attribute, providing a final evaluation score. This two-step process increases alignment with human grading and reduces the variability inherent in single-shot LLM evaluations.

3. Multi-Model Evaluation and Ensemble Scoring

Copernicus integrates outputs from multiple advanced language models (e.g., GPT-4, Claude, Gemini) to enhance the robustness and generalizability of evaluations. This

ensemble approach allows for cross-model comparison, detection of inconsistencies, and aggregation of scores for greater reliability.

4. Applications and Outputs

(i) The system has been applied to diverse contexts, such as evaluating scientific reasoning in corporate communication, comparing expert and novice scientific texts, and benchmarking organizational decision-making.

(ii) Results produced by Copernicus include fine-grained scores and detailed rationales for each attribute, enabling both quantitative analysis (e.g., correlating "scientificness" with organizational outcomes) and qualitative feedback for improvement.

5. Transparency and Extensibility

All rubrics, attribute definitions, and scoring code are transparent and modular, allowing for adaptation to new domains or constructs as required by future research or organizational needs. The system incorporates standard software engineering practices (e.g., version control, standardized JSON schemas) to ensure replicability and ease of extension.

Chapter 4

Pipeline Design

4.1 Overview of Pipeline Steps and Processes

The development of an external validation pipeline for benchmarking LLM evaluative performance centers on a sequence of carefully structured steps. The pipeline integrates survey delivery, participant sampling, rubric assignment, evidence collection, and automated as well as human scoring, to ensure reliable validation of LLM-based evaluation models.

1. **Sample Documents**

   (i) **Input:** Clean text(s); rubric and attributes.

   (ii) **Function:** Load and prepare sample documents for annotation; LLM parses content and associates it with the intended rubric.

   (iii) **Output:** Summarized evaluation criteria (JSON or Table).

   (iv) **Validation Criteria:** Evaluation criteria match document/rubric content and project objectives.

2. **Survey Content**

   (i) **Input:** Clean text(s); rubric and attribute list.

   (ii) **Function:** Insert markers and map attributes into survey structure, ensuring rubrics are reflected.

(iii) **Output:** Generated survey (JSON format).

(iv) **Validation Criteria:** Survey structure fits template and evaluation intent; all rubric elements are included.

3. **Validate Survey**

   (i) **Input:** Survey code; evaluation criteria.

   (ii) **Function:** LLM reviews and verifies survey logic, completeness, and clarity.

   (iii) **Output:** {`"validated": 1`} or {`"validated": 0`}

   (iv) **Validation Criteria:** Survey meets all design, logic, and clarity standards.

4. **Qualtrics Testing**

   (i) **Input:** Validated survey code; evaluation criteria.

   (ii) **Function:** Preview survey as rendered in Qualtrics and check for display or logic errors.

   (iii) **Output:** Qualtrics link (preview).

   (iv) **Validation Criteria:** Survey displays correctly; navigation and logic are functional.

5. **Prolific Filters**

   (i) **Input:** JSON of Prolific filters; evaluation criteria.

   (ii) **Function:** LLM selects and optimizes participant filters to align with survey goals.

   (iii) **Output:** Optimal filter set.

   (iv) **Validation Criteria:** Filters accurately represent the intended target sample and audience.

6. **Validate Prolific Filters**

   (i) **Input:** Filter set (JSON); evaluation criteria.

(ii) **Function:** Cross-check filters for logic, relevance, and alignment with recruitment objectives.

(iii) **Output:** {`"validated": 1`} or {`"validated": 0`}

(iv) **Validation Criteria:** Filter set meets demographic and survey needs; no logical contradictions or omissions.

7. **Prolific Recruitment**

   (i) **Input:** Validated filter set; survey link.

   (ii) **Function:** Launch survey to selected Prolific participants using defined filters.

   (iii) **Output:** Recruitment report (participant counts, demographics).

   (iv) **Validation Criteria:** Recruitment matches intended sample size and demographic profile.

8. **Data Collection**

   (i) **Input:** Active survey; Prolific participant pool.

   (ii) **Function:** Collect response data as participants complete survey.

   (iii) **Output:** Annotated data (CSV, JSON).

   (iv) **Validation Criteria:** Data completeness; no missing key responses; metadata (timestamps, participant ID) preserved.

9. **Data Validation**

   (i) **Input:** Raw response data.

   (ii) **Function:** Check data for quality (e.g., attention checks, completion status, time-on-task).

   (iii) **Output:** Cleaned dataset; validation report.

   (iv) **Validation Criteria:** All attention and logic checks passed; incomplete or invalid responses flagged or removed.

10. **Preprocessing and Analysis**

(i) **Input:** Cleaned dataset.

(ii) **Function:** Prepare data for analysis (normalization, formatting); generate summary statistics and initial analyses.

(iii) **Output:** Analysis-ready dataset; summary statistics.

(iv) **Validation Criteria:** Data is correctly formatted and ready for inferential/statistical analyses; reproducible processing scripts.

Table 4.1 summarizes each module of the pipeline, reflecting the structure in the project flow design.

Module	**Input**	**Function**	**Output**	**Validation Criteria**
Sample documents	1. Clean text(s) 2. Rubric and attributes	LLM reads samples and loads documents	Summarized evaluation criteria (JSON or Table)	Criteria reflect both documents and rubrics
Survey content	1. Clean text(s) 2. Rubric and attributes	Adds text markers, attributes, and builds survey	Generated survey (JSON)	Survey matches template; fits evaluation context
Validate survey	1. Survey code 2. Evaluation criteria	LLM validates survey logic and completeness	{"validated": 1} or {"validated": 0}	Meets logic, completeness, clarity standards
Qualtrics testing	1. Survey code 2. Evaluation criteria	LLM previews survey and checks logic/display	Qualtrics link	Correct display; preview functions as expected
Prolific filters	1. JSON of filters 2. Evaluation criteria	LLM selects optimal participant filters	Optimal filters	Filters fit survey requirements and audience
Validate prolific filters	1. JSON of Prolific filters 2. Evaluation criteria	1. Checks if the optimal filters are indeed in JSON 2. Re-validates against criteria	{"validated": 1} or {"validated": 0}	Prolific filters not in JSON, not optimal
Prolific Recruitment	1. Optimal Prolific filters 2. Prolific sample code	Modify the sample code to incorporate the selected filters and audience	Resulting updated code	Code is deployable and produces expected outcome
Prolific validation	1. Updated code 2. Evaluation criteria	Validate that prolific code works. If code does not work, regenerate	{"validated": 1} or {"validated": 0}	Recruitment script produces correct filtered link
Deploy	Updated code	Deploy survey via prolific using updated code	Prolific survey	
Monitor and log	Prolific	Continuously monitor the systems:Check survey launch, response rate, Prolific rejections	Store the logs with time stamps of when something fails	

Table 4.1: Stepwise overview of the pipeline modules, inputs, outputs, and validation criteria.

4.2 Trade-offs: Number of Rubrics, Items per Participant, Sampling Strategies

The pipeline design requires careful balancing of several competing factors, each with direct implications for data reliability, participant burden, and statistical power.

4.2.1 Number of Rubrics

Expanding the number of rubrics increases the granularity of evaluation and coverage of conceptual space, as emphasized by the HD-eval and Copernicus frameworks. However, more rubrics per artifact lead to higher cognitive load and survey fatigue among raters (Liu et al. 2024). Literature indicates that decomposing evaluation criteria into hierarchical or marker-based subcriteria improves reliability but must be weighed against practical feasibility (Boussioux et al. 2024; Zheng et al. 2023).

4.2.2 Items per Participant

Assigning more items per participant can reduce costs and maximize data yield but increases the risk of fatigue and satisficing. Sampling fewer items per participant allows for more focused and higher-quality ratings but necessitates recruiting more participants for the same total number of ratings. The design employs stratified sampling to balance these demands, distributing the assessment workload while ensuring each item receives multiple independent ratings. Additionally, assigning more items per participant will likely push them to use an LLM them self to validate the text.

4.2.3 Sampling Strategies

Diversity and scale are both crucial for robust aggregation, echoing findings from the "wisdom of crowds" literature and recent LLM benchmarking studies (Doshi et al. 2024). The pipeline adopts stratified random sampling over relevant demographic and behavioral filters, as enabled by Prolific's rich metadata (see prolific_filters.json), ensuring both rep-

resentativeness and diversity in perspectives. Additionally, randomization of text/rubric assignment helps minimize order effects and rater bias.

4.3 Design Decisions and Rationale

Key design choices are informed by both theoretical and practical considerations:

(i) **Hierarchical Rubric Decomposition:** Adopting hierarchical and marker-based rubric structures, following best practices from Copernicus and HD-eval, enables the measurement of nuanced evaluation criteria, supports evidence-based scoring, and aligns with current trends in LLM evaluation (Zheng et al. 2023).

(ii) **Two-Step Scoring (Markers Approach):** Implementing a two-step process (evidence extraction followed by holistic scoring) improves both validity and reliability, reducing stochasticity in LLM outputs and aligning scores more closely with human benchmarks .

(iii) **Assignment Algorithms:** The assignment of texts and rubrics is randomized, yet stratified to ensure each item is rated by a sufficient and diverse set of participants. This approach is motivated by findings that both diversity (across raters) and scale (number of ratings) improve aggregation accuracy and mitigate individual biases (Doshi et al. 2024).

(iv) **Quality Control Mechanisms:** Attention checks, duplicate items, and validation items are interleaved in the survey to detect inattentive or careless raters. Outlier detection and manual review are used for additional quality assurance.

(v) **Feedback and Iteration:** Regular reviews with team members inform iterative improvements to rubric definitions, survey structure, and assignment logic, as documented in team discussions and project retrospectives.

4.4 Addressing Quality and Reliability

Ensuring validity, quality, and reliability is fundamental to the pipeline's design:

(i) **Calibration and Training:** Human raters are provided with clear instructions, examples, and calibration rounds based on the defined rubric markers, reducing ambiguity and enhancing rubric fidelity.

(ii) **Automated Quality Checks:** Algorithmic detection of inconsistent, patterned, or low-effort responses is implemented. Participants failing quality thresholds are excluded from analyses.

(iii) **Comparison with LLM Outputs:** Human ratings are compared with LLM evaluations using the same pipeline, enabling external validation of model-based assessments and the identification of systematic biases or gaps.

4.5 Team Planning Considerations

The pipeline's evolution has been deeply informed by iterative team discussions and cross-functional feedback, particularly with methodological experts.

(i) **Rubric and Marker Iteration:** Frequent consultations addressed ambiguity in rubric definitions and the need for marker independence and comprehensive coverage. Input from domain experts led to clearer attribute and sub-concept delineation.

(ii) **Survey Burden and Participant Experience:** Discussions highlighted the necessity of minimizing participant fatigue, resulting in limits on the number of items per survey session and the inclusion of engagement checks.

(iii) **Assignment Logic:** Feedback on early assignment algorithms resulted in more robust stratified randomization, ensuring both fairness in workload and representativeness in sampling.

(iv) **LLM Reliability and Human Benchmarks:** Team debates emphasized the importance of aligning LLM scoring strategies (e.g., chain-of-thought prompting, rubric decomposition) with human annotation practices, to maximize comparability and reliability (Zheng et al. 2023).

(v) **Quality Control Policies:** Input from multiple stakeholders informed the development of transparent, replicable quality control standards, including the handling of inattentive raters and the use of gold-standard items for calibration.

Chapter 5

Qualtrics Implementation

5.1 Survey Development with Qualtrics

The development of the survey instrument within Qualtrics represented a foundational step in the data collection process for benchmarking Large Language Model (LLM) evaluative performance. Drawing upon best practices for experimental design and informed by requirements for external validation as articulated in prior chapters, the Qualtrics survey was structured to maximize clarity, respondent engagement, and alignment with the rubric-based evaluation criteria established for the study.

The survey was designed to elicit targeted responses that would serve as high-quality ground truth for subsequent benchmarking of LLM evaluative judgments. Informed by the Copernicus system's rubric decomposition methods, each item in the survey was mapped to specific attributes and markers, as defined in both the marker and subrubric definitions files . The use of precise, literature-based rubrics (as detailed in the Copernicus documentation and design guidelines) ensured that human responses could be reliably compared to LLM outputs on multiple granular dimensions of reasoning quality, such as articulation and justification of assumptions, problem formulation, attribute independence, and logical clarity .

Survey development also took into account the need for systematic participant filtering and demographic diversity, leveraging insights from participant filter specifications

(see prolific_filters.json) to construct branching logic that appropriately routed participants based on eligibility and demographic variables. This enabled the collection of a stratified and representative dataset, further supporting the validity of downstream LLM evaluation.

5.2 Use of the Qualtrics API

Automation of the survey deployment, monitoring, and data export processes was accomplished through the use of the Qualtrics API, as detailed in the associated implementation notebook `Qualtrics_w_Prolific.ipynb`. The API enabled programmatic creation and modification of survey instruments, batch import of question blocks mapped from Copernicus rubric items, and seamless integration with external platforms such as Prolific for participant recruitment and tracking.

Through the API, survey flow and logic were dynamically adjusted in response to pilot feedback and rubric refinements, supporting an iterative, version-controlled survey development process. The integration of the Qualtrics API also streamlined post-collection data handling, automating the extraction of structured responses in formats suitable for rubric-based coding and subsequent LLM benchmarking. Automation scripts facilitated the mapping of participant responses to evaluation pipelines, including the assignment of responses to human and LLM evaluators under blinded conditions.

5.3 Key Steps of Survey Implementation

The implementation of the Qualtrics survey followed a systematic, multi-stage workflow:

1. **Rubric Mapping and Question Generation:** Each survey item was explicitly mapped to corresponding rubric criteria and subrubrics defined within the Copernicus measurement model, ensuring that human raters and LLMs would be evaluated against a consistent set of standards .

2. **Participant Screening:** Demographic and eligibility filters, derived from the pro-

lific_filters.json file, were implemented as screener questions and branching logic to ensure that only appropriate respondents proceeded to the main survey blocks. This maximized the relevance and generalizability of the dataset .

3. **API-Based Deployment and Monitoring:** The Qualtrics API was used to automate survey publishing, participant invitation, and real-time monitoring of response rates and completion metrics .

4. **Data Extraction and Formatting:** Upon survey completion, response data were automatically exported via the API in standardized formats compatible with downstream rubric-based evaluation and LLM benchmarking workflows.

5. **Quality Assurance:** Automated and manual checks were applied to detect inattentive responses, speeding, or inconsistent answers. Only high-quality, attention-checked data proceeded to the benchmarking phase.

Throughout the implementation, survey design choices were made with reference to both methodological rigor and practical constraints, such as sample size, respondent incentives, and survey length. The Qualtrics-based workflow ensured traceability between human and LLM evaluation outputs, forming a robust empirical foundation for the thesis's subsequent benchmarking analyses.

Chapter 6

Prolific Study Launch

6.1 Integrating Prolific and Qualtrics

This section details the integration of Prolific, a leading online participant recruitment platform, with Qualtrics, the primary tool used for survey administration and data collection in this thesis. The integration of these systems is critical for enabling the scalable, automatic, and reliable collection of human evaluation data necessary for benchmarking large language model (LLM) evaluative performance.

The Copernicus pipeline employs a streamlined data flow wherein participants are first recruited and screened via Prolific, then routed into the Qualtrics survey environment, and subsequently have their survey completions and metadata communicated back to Prolific. This integration ensures that participant eligibility, response tracking, and compensation can be managed programmatically, thereby supporting high throughput and robust data quality control.

The integration process typically involves three primary components:

1. **Participant Recruitment and Filtering:** Prolific's robust filtering infrastructure enables targeting of specific demographic or experiential attributes (see Section 6.2). Recruitment links are generated in Prolific and customized with participant IDs that are passed as embedded data fields to Qualtrics.

2. **Survey Access and Completion Tracking:** Upon accessing the Qualtrics sur-

vey, participants' IDs and relevant metadata are automatically recorded. Custom survey logic ensures that only eligible, pre-screened participants can proceed, and attention checks or quality controls can be applied in real time.

3. **Completion Verification and Automated Compensation:** After survey submission, Qualtrics redirects participants back to Prolific, appending unique completion codes or status flags that facilitate automatic approval, bonus calculation, or rejection based on pre-specified criteria. This closed-loop system is essential for preventing fraud and ensuring timely remuneration.

The pipeline leverages Qualtrics' API and Prolific's webhook features to enable seamless transfer of status updates, survey outcomes, and participant-specific data. This programmatic integration is particularly important for large-scale, multi-condition studies that characterize modern LLM evaluation pipelines, reducing manual effort and minimizing the risk of human error in participant handling.

6.2 Participant Criteria, Remuneration, Demographics

Defining participant criteria is a foundational element in the design and launch of a Prolific-based study. The credibility and external validity of LLM evaluation benchmarks depend substantially on the demographic and experiential diversity of the recruited sample, as well as the application of appropriate remuneration schemes.

6.2.1 Participant Criteria

Participant criteria are established using Prolific's advanced filtering mechanisms, informed by the JSON of available demographic and behavioral attributes. These include, but are not limited to, age, gender, educational attainment, employment status, health status, and a wide range of political, economic, and lifestyle factors. The platform enables complex eligibility screening by combining multiple filters—for instance, selecting

only those in a particular country, education level, or with previous experience in relevant domains. Filters are documented, regularly updated, and extensible, ensuring that each study targets its intended population effectively.

6.2.2 Remuneration

Remuneration is calibrated according to Prolific's best practice guidelines, which aim to ensure ethical compensation while maximizing recruitment efficiency and data quality. Rates are set at 10 Pounds (GBP) an hour.

6.2.3 Demographic Balancing and Reporting

Thorough documentation of the recruited sample's demographics is required to support reproducibility and future meta-analyses. Data collected includes standard variables (age, gender, country of residence, education, etc.), and can be further enriched with contextual information relevant to the study (such as prior AI experience or familiarity with specific task domains). This documentation not only enables subgroup analyses, but also ensures that the evaluation pipeline's findings can be interpreted in light of potential sampling biases.

6.3 Monitoring and Data Handling

Rigorous monitoring and data handling are essential to uphold the scientific integrity of studies involving human participants, particularly when such studies are foundational for the external validation of LLM evaluative performance.

6.3.1 Data Security and Privacy

All participant data are handled in accordance with institutional ethics protocols and GDPR requirements. Prolific and Qualtrics both implement secure data transfer and storage practices; personally identifying information (PII) is separated from survey response data, and access is restricted to authorized personnel.

6.3.2 Quality Control and Exclusion Criteria

Post-hoc data quality assessment involves screening for incomplete responses, failed attention checks, or evidence of fraudulent participation (e.g., using LLMs to evaluate the text instead of doing so manually). The pipeline enables automated flagging and exclusion of such responses, with logs maintained for auditability. Participants who fail pre-specified quality control checks are automatically notified and, where appropriate, not compensated for their participation.

6.3.3 Data Export and Integration

Finalized, high-quality response data are exported from Qualtrics in standardized formats (e.g., CSV or JSON) and linked with Prolific's participant metadata. This enables downstream integration with the evaluation and analysis pipelines, supporting both human- and LLM-based scoring of responses. The integrity and traceability of data from recruitment through to analysis is thus preserved.

Chapter 7

Evaluation Methods and (AI) Agents

7.1 Role of LLMs and AI Agents for Evaluation

Large language models (LLMs) and AI agents are increasingly central to automated evaluation frameworks, serving both as scalable judges and as sources of synthetic evaluation data. The shift toward "LLM-as-a-judge" or "Language-Model-as-an-Examiner" paradigms reflects both the need for scalable assessment and the desire to approximate human preferences in evaluation tasks where direct human annotation is costly or impractical (Gu et al. 2024; Zheng et al. 2023). LLMs can be prompted to rate responses, simulate Likert-scale scoring, or even emulate specific evaluative roles, enabling their deployment across diverse benchmarking, survey, and validation pipelines.

The Copernicus system exemplifies such agentic evaluation, leveraging state-of-the-art LLMs (e.g., GPT-4o, Claude, Gemini) for scoring text on abstract constructs like scientific reasoning, customer centricity, or creativity . These models are integrated into multi-step pipelines, where their responses are not just "one-shot" ratings but the result of structured evidence extraction and rubric-based reasoning. This approach has demonstrated better alignment with human evaluators than naive LLM scoring, particularly when combined with techniques such as chain-of-thought prompting, hierarchical rubric decomposition,

and evidence aggregation (Zheng et al. 2023).

Beyond scoring, LLM agents are now being tasked with the simulation of survey respondents (e.g., for synthetic data generation), the emulation of behavioral responses in social science experiments, and even the modeling of complex agent-based interactions (Jiang et al. 2025; Park et al. 2024). In these roles, LLMs must not only reproduce mean patterns of human judgment, but also capture population-level diversity, intra-respondent variability, and sensitivity to context.

Recent research has highlighted both the promise and limitations of LLMs in these evaluative roles. For example, LLMs can approximate aggregate survey results and capture cross-cultural differences, but may also exhibit systematic biases or over-homogenize responses, especially on sensitive or value-laden topics (Jiang et al. 2025; Park et al. 2024). In creative and strategic domains, human-AI collaboration—where LLMs are guided by human prompt engineering or used in the loop for idea generation and evaluation—can yield solutions with higher practical value and strategic viability than human or AI-alone baselines (Boussioux et al. 2024).

7.2 Methodology for Evaluation/Validation

7.2.1 External and Human Validation

Validation of LLM-based evaluation is performed by comparison with human expert ratings, external datasets, or behavioral ground truths. Systematic benchmarking studies (e.g., on business model evaluations, creative problem solving, or agent-based social simulation) compare LLM scores with those of human judges, assess correlations, and test generalizability across contexts (Boussioux et al. 2024; Doshi et al. 2024; Park et al. 2024). In Copernicus, human-scored reference sets are used for calibration and to assess system accuracy.

7.3 Reliability and Validity Considerations

7.3.1 Reliability

Reliability in the context of LLM evaluative performance refers to the consistency of model outputs under repeated or varied conditions. Key considerations include:

(i) **Stability Across Runs**: Consistency of model ratings when prompts are repeated with the same or randomized seeds.

(ii) **Sensitivity to Prompting**: Variation in output due to changes in prompt phrasing, rubric structure, or example selection.

(iii) **Aggregation Effects**: The extent to which consensus methods (e.g., averaging multiple model runs) increase reliability (Doshi et al. 2024; Zheng et al. 2023).

Recent work demonstrates that reliability is not universal; it can depend on the LLM model, evaluation task, temperature settings, and other hyperparameters (Zheng et al. 2023). The Copernicus pipeline explicitly addresses reliability by using multi-step evaluation, prompt templates, and aggregation over multiple runs.

7.3.2 Validity

Validity concerns whether the LLM evaluation accurately reflects the construct or judgment intended. Several aspects are addressed:

(i) **Construct Validity**: Ensured by rigorous definition and operationalization of attributes and markers, grounded in literature and expert review .

(ii) **Criterion Validity**: Assessed by comparing LLM-derived scores to human expert judgments and to real-world outcomes (e.g., venture survival, competition results) (Doshi et al. 2024).

(iii) **Bias and Calibration**: Systematic differences between LLM and human responses—due to model pretraining, cultural bias, or over-homogenization—are identified and mit-

igated through prompt design and validation studies (Jiang et al. 2025; Park et al. 2024).

(iv) **Generalizability**: Testing across diverse tasks and domains (e.g., surveys, strategic decision making, creativity) supports validity beyond narrow contexts (Boussioux et al. 2024; Park et al. 2024).

7.3.3 Current Challenges and Opportunities

While LLM evaluators show strong potential for scalability and cost reduction, current limitations remain:

(i) **Stochasticity and Instability**: Single model outputs can vary; aggregation and prompt engineering are needed for robustness.

(ii) **Sensitivity to Context**: LLM judgments may be influenced by subtle prompt variations or by the context embedded in training data.

(iii) **Cultural and Social Bias**: LLMs may exhibit systematic bias toward certain viewpoints or normative positions, particularly on sensitive or value-laden issues.

(iv) **Limitations in Novelty and Creativity**: LLMs can homogenize responses or struggle with tasks requiring extreme novelty, although human-AI collaboration can improve results in creative domains (Boussioux et al. 2024).

(v) **Benchmarking and Human Alignment**: Continued comparison with human raters, especially in high-uncertainty or low-consensus tasks, is needed to ensure meaningful alignment.

Chapter 8

Discussion

8.1 Implications of the Workflow Approach

The workflow approach developed and benchmarked in this thesis—centering on the Copernicus pipeline for automatic LLM validation—presents several significant implications for both research and practice in the evaluation of large language models (LLMs).

First, the agentic, rubric-based workflow enables scalable, systematic, and transparent evaluation of complex, abstract constructs in text, such as scientific reasoning and creativity. By breaking down high-level evaluation criteria into granular, well-defined "markers" and subrubrics (as detailed in), the Copernicus pipeline supports not only the reproducibility of evaluations but also enhances their interpretability and alignment with human reasoning standards. This two-step "evidence extraction and evaluation" process, where LLMs first generate detailed explanations for each marker and then synthesize these into holistic attribute scores, offers improved accuracy and reliability over standard one-shot LLM assessment techniques.

Second, the Copernicus workflow's integration of multiple LLMs (e.g., GPT-4o, Claude-Sonnet, Gemini Pro) as evaluators leverages diversity and scale—a principle consistently highlighted in recent literature as key to robust model judgment and prediction. Aggregating results from heterogeneous models, prompts, and agent "roles" (such as expert or layperson perspectives) systematically improves alignment with human benchmarks by

offsetting model-specific biases and reducing individual evaluator inconsistency.

Third, this workflow addresses longstanding challenges in LLM evaluation identified by recent research, including the limitations of internal validation, the need for human-comparable external benchmarks, and the stochasticity inherent to LLM outputs. By operationalizing hierarchical rubric decomposition, chain-of-thought reasoning, and structured prompt engineering within the pipeline, the Copernicus approach positions itself as a state-of-the-art solution for practical LLM benchmarking.

In sum, the workflow enables:

(i) Efficient, transparent, and replicable model evaluation across a range of domains.

(ii) Empirical grounding for claims about model "scientificness" or reasoning ability, supporting research on the role of artificial evaluators in complex decision-making.

(iii) The foundation for further integration of agent-based simulations and synthetic data generation, as demonstrated in recent work on generative agent banks and population-level modeling.

8.2 Limitations

Despite these contributions, the workflow developed in this thesis has several limitations that merit careful consideration:

8.2.1 Prompt and Hyperparameter Sensitivity

Evaluation performance remains sensitive to prompt design, chain-of-thought exemplars. Studies reviewed in this thesis demonstrate that the optimal settings for reliable model judgment are highly task- and model-dependent, and that performance may degrade if prompts are not carefully tailored or diversified. While the Copernicus workflow mitigates some of these issues via aggregation, this does not fully resolve prompt sensitivity as a source of potential unreliability.

8.2.2 Generalization to New Domains

Although the Copernicus system is demonstrated on attributes such as scientific reasoning, its applicability to domains with more ambiguous, subjective, or less well-understood criteria (e.g., creativity, ethics) may be limited. Attributes that lack consensus or objective ground truth pose challenges for both rubric construction and LLM-based evaluation, as highlighted by recent literature on creative text assessment.

8.2.3 Human-Likeness vs. Human-Benchmarking

Recent advances in generative agent simulations and LLM-based survey proxies reveal that while models can achieve high average agreement with human responses, they may still fail to reproduce the full diversity, biases, and unpredictability of real populations. This underscores a fundamental limitation: LLMs are best viewed as "approximators" of human judgment, rather than replacements for genuine human evaluation, especially in sensitive, value-laden, or high-stakes contexts.

8.3 Recommendations and Considerations for Future Work

Building on these findings and limitations, several avenues for future research and development emerge:

8.3.1 Dynamic and Adaptive Rubric Construction

Rubric construction should be an iterative, evidence-based process, involving not only domain expert input but also empirical validation against new data and model outputs. Approaches such as hierarchical criteria decomposition and human-in-the-loop refinement, as outlined in the literature and internal guides, are likely to improve both coverage and clarity.

8.3.2 Expanding Evaluator Diversity and Agentic Roles

Future versions of the Copernicus workflow should further expand the diversity of evaluator models, roles, and perspectives—potentially incorporating agents conditioned on different demographic profiles, expertises, or reasoning styles. Aggregation strategies that balance scale with diversity (e.g., combining outputs from multiple model types, prompt templates, and role-played evaluators) have been shown to systematically enhance alignment with human benchmarks and mitigate model bias.

8.3.3 Integration with Human Benchmarking and Feedback Loops

To better anchor automated evaluations in genuine human standards, closer integration with human benchmarking studies is recommended. Feedback loops wherein human experts review, correct, or calibrate LLM outputs—especially in ambiguous or low-consensus domains—can improve both model and rubric reliability. Additionally, joint analysis of variance, confidence calibration, and demographic parity differences should be used to identify and reduce biases.

8.3.4 Generalization to Open-Ended and Creative Tasks

The Copernicus approach should be extended to open-ended, creative, or less-structured evaluation settings, as highlighted by recent work on LLMs for creativity and innovation assessment. Here, new markers and rubrics may need to be developed, and the limits of current LLM capabilities for subjective evaluation rigorously tested.

8.3.5 Transparency, Open Science, and Benchmark Sharing

Transparency in rubric definitions, marker logic, and model selection is essential for reproducibility and trust in automatic evaluation systems. Open sharing of evaluation scripts, annotated datasets, and agent benchmarks—modeled after agent banks and open science initiatives—will support broader adoption and collective advancement.

8.3.6 Continued Assessment of Reliability, Validity, and Bias

Ongoing work should empirically assess the reliability (consistency), validity (accuracy), and fairness (absence of systematic bias) of both the evaluation pipeline and its LLM components. Attention should be paid to context drift, prompt artifacts, and differences in performance across demographic groups, domains, and task types.

Chapter 9

Conclusion

9.1 Summary of Contributions

This thesis, *Benchmarking LLM Evaluative Performance: Development of a Pipeline for Automatic Validation*, has advanced the field of AI-driven evaluation by designing, implementing, and empirically validating a novel workflow for benchmarking large language models (LLMs) as evaluators. The work addresses a critical gap in the automation and external validation of evaluative performance, especially for abstract, complex constructs frequently encountered in research, industry, and policy contexts.

The primary contributions of this work are as follows:

(i) **Development of the Copernicus System:** The thesis introduced Copernicus, a language-based measurement tool enabling systematic, rubric-driven evaluation of difficult-to-measure concepts—such as "scientificness" or "customer centricity"—directly from textual artifacts. Copernicus integrates agentic AI to generate, customize, and deploy measurement models at scale, enabling both fine-grained and high-throughput evaluation in natural language settings. The system was designed to address the need for reliable, externally valid metrics for scientific reasoning and related constructs, providing both theoretical grounding and practical relevance for organizational and academic users.

(ii) **Two-Step Markers-Based Evaluation Pipeline:** The core innovation of Coper-

nicus is a two-step evaluation protocol using granular "markers"—well-defined sub-criteria aligned with theoretical constructs—and detailed rubrics to drive both evidence extraction and evidence evaluation. This approach, as demonstrated in practical use cases, increases the reliability, granularity, and validity of LLM evaluations compared to traditional one-shot scoring approaches. By combining evidence extraction (step 1) with structured evidence evaluation (step 2), Copernicus ensures that LLM-based assessments are both interpretable and benchmarked against human expert ratings.

(iii) **Integration with External Validation Pipelines:** The thesis details the integration of Copernicus into an automated validation pipeline using survey platforms (Qualtrics), participant recruitment (Prolific), and agentic evaluation. By connecting human and LLM-based assessments, the workflow supports robust benchmarking and quality assurance, enabling quantitative assessment of LLM reliability, validity, and calibration across diverse domains and problem types.

(iv) **Advancement in Rubric and Marker Methodology:** The work provides detailed methodological contributions for rubric and marker design, drawing on best practices from educational measurement, the Copernicus project's own internal definitions, and current literature on LLM-based evaluation. The thesis introduces and operationalizes a set of granular attributes (e.g., Assumptions, Problem Formulation, Attribute Formulation, Logical Links, Data Orientation, Belief Update, Methodic Doubt) and demonstrates their use in both system implementation and empirical evaluation.

(v) **Empirical Benchmarking of LLM-as-a-Judge Paradigm:** Through integration with external datasets and large-scale validation studies, the thesis provides empirical evidence for the reliability and limitations of LLMs as evaluators and judges. Key findings include the importance of prompt engineering (e.g., few-shot and rubric-based prompting), the effect of aggregation ("wisdom of crowds" effects), and the need for careful consideration of reliability, calibration, and potential bi-

ases in LLM outputs. Comparative studies with human experts highlight both the promise and the current limits of LLM evaluation in strategic, creative, and survey-based contexts.

(vi) **Best Practices for External Validation in AI Evaluation Pipelines:** The thesis synthesizes lessons learned regarding sampling, survey instrument design, data quality assurance, and the integration of human and LLM scoring for robust external validation. Emphasis is placed on transparent rubric design, use of granular sub-criteria (markers), explicit handling of uncertainty and bias, and the use of large-scale participant data for benchmarking AI performance.

Taken together, these contributions provide a reproducible, extensible framework for benchmarking LLM evaluative performance—bridging the gap between theoretical measurement, practical implementation, and empirical validation.

9.2 Final Remarks

The development and validation of the Copernicus pipeline mark a significant advance in the automation and standardization of evaluative processes for large language models. By grounding the approach in theoretically informed, rubric-based frameworks and leveraging the strengths of agentic AI, this thesis lays the groundwork for future research and applications in automated evaluation, scientific reasoning assessment, and beyond.

Several insights emerge from the work:

(i) **Scalability and Cost-Effectiveness:** The integration of agentic LLMs within automated validation pipelines enables scalable evaluation across a wide range of domains, reducing the cost and effort of human evaluation while maintaining—or even enhancing—consistency and reproducibility.

(ii) **Limitations and Ongoing Challenges:** While LLMs can approximate human evaluators in many settings, persistent challenges remain—particularly in capturing subjective, culturally contingent, or highly creative aspects of evaluation. As

evidenced by comparative studies, LLMs may exhibit systematic biases or reduced variance compared to human raters, necessitating careful external validation and aggregation strategies.

(iii) **Importance of Rubric and Marker Granularity:** Detailed, theoretically-grounded rubrics and sub-criteria are critical for guiding LLM evaluation, ensuring alignment with expert judgment, and enabling meaningful diagnostic feedback. This marker-based approach is broadly applicable and offers a template for further innovation in AI evaluation design.

(iv) **Human-AI Collaboration and the Path Forward:** The thesis highlights the complementarity of human and AI evaluation. Human expertise remains vital for rubric creation, edge-case adjudication, and continual refinement of measurement models. Future advances in AI evaluation will likely involve increasingly sophisticated human-AI collaboration, including adaptive pipelines, iterative prompt engineering, and integration with new modalities and evaluation tasks.